

CORS

CORS stands for **Cross-Origin Resource Sharing**.

It is a **browser security mechanism** that controls whether a website can make requests to a server on a **different origin**.

◆ What is an Origin?

An **origin** is made of:

- **Protocol** → http / https
- **Domain** → example.com
- **Port** → 3000, 5000

Example:

- http://localhost:3000
- http://localhost:5000

These are **different origins** because the ports are different.

◆ Why CORS Exists?

Browsers block many cross-origin requests by default for security.

Without CORS, malicious websites could access private data from another site.

CORS lets the **server decide** which origins are allowed.

◆ Example in MERN Stack

- React frontend → http://localhost:3000
- Express backend → http://localhost:5000

When React tries to fetch data from backend, browser blocks it unless backend enables CORS.

◆ Express.js Example

Install:

```
npm install cors
```

Use in server:

```
const express = require("express");
const cors = require("cors");
```

```
const app = express();
```

```
app.use(cors());
```

```
app.listen(5000);
```

This allows requests from all origins.

◆ Allow Specific Frontend Only

```
app.use(cors({
  origin: "http://localhost:3000",
  credentials: true
}));
```

This allows only your React app.

◆ Important CORS Headers

Server sends headers like:

- Access-Control-Allow-Origin
 - Access-Control-Allow-Methods
 - Access-Control-Allow-Headers
-

◆ What is Preflight Request?

For some requests, browser first sends an **OPTIONS** request to ask permission before the real request.